Runaldo Borah

me25mtech11023@iith.ac.in | ronaldo34borah@gmail.com | +91 8486045092 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.84
Gurukul Junior College	Class XII (AHSEC)	2020	58.60%
Tengakhat Jatiya Vidyalaya	Class X (SEBA)	2018	73.33%

RESEARCH EXPERIENCE

Hokkaido University, Graduate School of Engineering - Sapporo, Japan

Research Intern, Energy and Environmental Systems (JICA-funded)

- Performed ultrasonic velocity Profiling(UVP) experiments on shear thinning fluid shear-thinning pipe flows of Xanthan Gum and Polyacrylamide (PAM) solutions, generating the validation dataset for the M.Tech thesis.

WORK EXPERIENCE

Tata Technologies - Assam Government ITI Upgrade Project (via Sadguru Services) - Assam, India

Consultant - Subject Matter Expert (SME)

- Installed, commissioned and tested capital training equipment at Government ITIs and polytechnics across Assam.
- Completed industrial training in Electric Vehicle (EV) technology and Industrial Robotics.

PROJECTS

Hybrid Real-Simulation of Non-Newtonian Fluid Flow using ultrasonic velocity profiler

M.tech Thesis

- Developing a hybrid numerical-experimental framework in which a CFD solver for non-Newtonian pipe flow is nudged (Newtonian relaxation) toward Ultrasonic Velocity Profiling (UVP) measurements, so the simulated flow field stays consistent with the real experiment

Developing Heat Transfer in a Ribbed Square Channel

Course Project - Turbulence, IIT Hyderabad

- Investigated the effect of different rib angles on heat transfer and pressure drop in a ribbed square channel through CFD simulations using **ANSYS Fluent**, and compared the thermal performance of different rib configurations.

Design and Fabrication of a Solar Operated Vapour Absorption Refrigeration System

B.tech Project

- Designed, fabricated and experimentally evaluated a solar-operated vapour absorption refrigeration system to study the feasibility of renewable energy-based cooling applications.

INTERNSHIPS

Oil India Limited - Duliajan, Assam

Summer Intern, OGPS Department

- Gained field exposure to well completion and intervention, workover operations and secondary recovery in an onshore oilfield.

- Studied production equipment, utility systems and industrial safety procedures.

Dibrugarh University, Department of Petroleum Technology - Dibrugarh, Assam

Academic Intern

- Characterised crude oil by viscosity, density, distillation and impurity analysis.

POSITIONS OF RESPONSIBILITY

Member | Vibe Music Club, IIT Hyderabad

- Participated in club activities and collaborated in organizing cultural events.

Founder and Administrator | Golaghat Engineering Confessions (Facebook page)

- Built and managed an online student community for 3+ years, leading a team of page admins.

SKILLS

CAD/CAE Softwares

Ansys fluent, ICEM CFD, Star CCM+,

Programming languages

- MATLAB, Python, C

RELEVANT COURSEWORK

Advanced Heat Transfer, Incompressible Fluid Flow, Computational Fluid Dynamics, Advanced CFD, Turbulence, Combustion Technology, Machine Learning and its application